



# Anton Vikharev

Unity Senior Engineer · AI-Assisted Engineer

## CONTACT

### LOCATION

Tbilisi, Georgia

### CONTRACTING

Registered company and bank account in Tbilisi, Georgia – invoicing in USD / EUR

### HOURS

UTC+4 · 7 years working with US Pacific teams

### EMAIL

chaincastsystems@gmail.com

### LINKEDIN

linkedin.com/in/antivortex

### GITHUB

github.com/Antivortex

### BLOG

medium.com/@chaincastsystems

## SKILLS

### Engine & Languages

Unity, C#, .NET, C++, Python

### Multiplayer & Simulation

client-side prediction, latency compensation, deterministic physics, authoritative servers, Photon (PUN + server plugins), ECS / DOTS, cheat detection

### Gameplay & AI

behaviour trees, A\* pathfinding, procedural generation, custom shaders, Spine, DOTween, game-design math tooling

### Performance & Tooling

Unity Profiler, build-size and memory optimization, Addressables and AssetBundles, custom Editor tooling, Zenject, Git

### AI-Assisted Engineering

Claude Code, multi-agent orchestration, spec-driven development, automated code review and regression gates

## PROFESSIONAL SUMMARY

Unity gameplay and systems engineer with 12 years in software development, 7 of them remote with a US studio in Menlo Park, CA. Shipped 4 commercial games – including Travel Crush, 1M+ downloads – and a cross-platform Unity SDK for a SaaS platform. Led teams of up to 4 engineers, ran technical interviews, and owned features from product requirement through architecture to shipped code.

Deep in the parts of Unity that are hard to hire for: deterministic simulation and custom 2D physics, client-side prediction netcode, behaviour-tree and A\* game AI, build-size and framerate optimization, Addressables delivery pipelines (including implementation of custom ones, and live-ops content systems.

I have years of experience and supporting gameplay, social and leaderboard backend systems with Python, Docker, AWS, PostgreSQL, Redis and .NET.

I now build agent-first. I orchestrate Claude Code agents under spec-driven workflows with automated review and regression gates – currently leading an AI-native CRM in Next.js / Express / PostgreSQL / Docker, delivered by a team of 3 junior engineers working alongside agents. I apply the same workflow to Unity tooling, prototyping and gameplay systems.

## WORK EXPERIENCE

### AI Assisted Engineer

March 2026 – Present

#### TutorEge – Remote

*Production CRM for an operating offline education business*

- Lead an AI-native build of a production CRM, from stakeholder discovery through delivery
- Orchestrate Claude Code agents to implement the system in Next.js / Express / PostgreSQL / Docker, using spec-first workflows with automated review and regression gates
- Ran discovery with end users and stakeholders to design UI/UX around existing business processes rather than forcing the business to adapt to the software

### Independent Unity & Backend Engineer

August 2024 – March 2026

#### Own practice – contract engineering – Remote

- Prototyped an MMO architecture sustaining 1,000 concurrent players on a custom ECS backend server with a Unity ECS client
- Delivered a match-3 prototype in Unity for the Banana Games startup, from empty project to playable vertical slice
- Built out an AI-assisted engineering practice: agent-driven refactors, test generation and tooling for C# / Unity codebases
- Discovered high-severity vulnerability in UMA Protocol (Polymarket's oracle) Solidity smart contracts (audited by top security firms prior to finding), earning a \$10,000 bug bounty

### BebopBee, Inc.

July 2016 – August 2024 · 7 years, three tenures

#### Menlo Park, CA, USA – fully remote

#### Unity Tech Lead

August 2023 – August 2024

*Travel Crush – match-3 mobile game, 1M+ downloads*

- Led a team of 4 client engineers; owned architecture and implementation plans for client-server features alongside the product team
- Migrated the game backend from Python / Tornado to BrainCloud BaaS
- Shipped multiple live-ops events on the no-code content pipeline
- Built a procedural animation system letting effects artists author dynamic flight curves for UI elements without engineering time
- Ran technical interviews for engineering candidates and drove sprint planning, translating product requirements into data structures, algorithms and technical specifications

## Backend

ASP.NET Core, EF Core, PostgreSQL, Redis, RabbitMQ, Node / Express, Docker, BrainCloud BaaS

## LANGUAGES

English – fluent (7 years in US engineering teams)

Russian – native

## EDUCATION

September 2007 – June 2013

**Bauman Moscow State Technical University**

MS in Ballistics and Aerodynamics

Thesis on optimal out-of-plane orbital transfer, LEO to GEO, via Lambert's problem

## Unity / C# / Python Fullstack Engineer

March 2018 – August 2022

*Travel Crush*

- Built the core match-3 engine and an in-editor level designer that let designers play and tune levels without leaving Unity
- Created a dynamic difficulty system that auto-balances level difficulty against player skill to sustain engagement
- Developed AI-driven automated level playtesting for bug detection and difficulty analytics
- Engineered a live-ops event system enabling no-code content delivery
- Built a custom Addressables system for grouping and background-downloading asset bundles
- Optimized UI for high-framerate rendering, including embedded scroller views, and wrote custom shaders for sprite rendering
- Implemented social systems in the Unity client and Python backend: friends graph, social notifications, global and in-event leaderboards on Redis, player-to-player gifts
- Integrated Facebook, Google and Apple sign-in across client and server, plus an embedded third-party video player bringing video content into the game meta

## Unity / C# Software Engineer

July 2016 – March 2018

*Super Jump League – real-time multiplayer PvP*

- Built a deterministic 2D physics engine (quad-to-quad, quad-to-tilemap, quad-to-circle collisions) as the basis for reproducible networked gameplay
- Implemented the networking engine with client-side prediction and latency compensation for smooth synchronization between clients in session multiplayer
- Implemented A\* pathfinding with behaviour trees for bot AI, fully synchronized across clients
- Built cheat detection via Photon server plugins
- Designed the session multiplayer lobby, with bots backfilling low-population hours
- Designed and implemented the character progression meta-game system
- Cut build size by 50% in one week; drove a bug-free soft launch through systematic QA and polish

## Unity / C# Software Engineer

August 2022 – June 2023

**Ankr – Remote**

*Cross-platform SDK for a Web3 SaaS platform*

- Sole Unity engineer – shipped a cross-platform Unity SDK, including WebGL, one month ahead of platform launch
- Integrated third-party APIs for cross-platform transaction processing inside the SDK
- Contributed to ASP.NET backend services on PostgreSQL, EntityFrameworkCore and RabbitMQ after SDK delivery

## Unity / C# Software Engineer

August 2015 – June 2016

**Crazy Panda – Moscow**

*Stellar Age – mobile MMO strategy*

- Created procedural galaxy generation for 100,000+ planets
- Built a quad-tree rendering system for smooth infinite zoom from galaxy view to planet view
- Built Editor extensions for validating game-design math

## C++ Software Engineer

May 2014 – December 2014

**Biart – Moscow**

- Gameplay and engine work on Depth Hunter 2: Deep Dive, released on Steam